Arnav Goyal

+91-88516-97733 goyalagf@gmail.com Linkedin
GitHub LeetCode CodeChef Codeforces

Summary

Computer Science graduate with hands-on experience spanning software engineering, cybersecurity, and embedded/automotive security. Built full-stack applications and conducted security assessments across embedded systems, wireless protocols, and web/mobile platforms.

Education

Jaypee Institute Of Information Technology **July 2023 – May 2026**
*Bachelor of Technology in Computer Science & Engineering (Lateral Entry); CGPA: **8.9/10.0*** *Noida, Uttar Pradesh*

- Qualified **GATE CSE 2026**, securing an **AIR 8527**.

Vellore Institute of Technology **September 2022 – July 2023**
*Bachelor of Technology in Computer Science & Engineering; CGPA: **8.4/10.0*** *Vellore, Tamil Nadu*

Amity International School **April 2018 – May 2022**
*High School; Class 12th CBSE: **96.6%** (2022), Class 10th CBSE: **96.2%** (2020)* *Noida, Uttar Pradesh*

- Ranked among the **Top 10 students out of 500+** academically in Class 9th, reflecting consistent high performance.
- Served as a **Scribe** for a fellow student with special needs during **Class 10th Board Exams**, while in Class 9th.

Experience

Thales Group **January 2026 – August 2026**
Software Engineering Intern – Embedded Security *Noida, Uttar Pradesh*

- Conducted **embedded hardware penetration testing** on automotive **ECUs** using **JTAG**, **SWD**, **OpenOCD**, and **GDB** to inspect memory, analyze firmware, and assess hardware debug interface security.
- Performed security assessment of automotive communication protocols through **CAN bus sniffing**, **message replay/injection**, and **UDS diagnostic service** analysis to evaluate ECU attack surfaces and identify potential security vulnerabilities.
- Executed **Bluetooth Low Energy (BLE)** security testing using **nRF52840**, **Ubertooth One**, **Wireshark**, and an **RF-shielded (Faraday Cage)** environment, analyzing **GAP/GATT** communication, advertising packets, pairing procedures, and connection establishment.
- Conducted **Android application security assessments** through **static** and **dynamic analysis** using **JADX**, **JD-GUI**, **APKTool**, **Drozer**, and **ADB**, performing **reverse engineering** and vulnerability analysis on both intentionally vulnerable and production Android applications.

Fluidech **June 2024 – July 2024**
Cyber Security Intern *Gurugram, Haryana*

- Executed vulnerability assessments and penetration tests on web applications using **OWASP ZAP, Burp Suite, and Nmap** to identify and report critical security flaws.
- Managed real-time incident response and threat detection utilizing **Wazuh SIEM** for log analysis, reinforcing data protection through **cryptographic and networking protocols**.

Projects

Real-Time Bus Ticketing System **GitHub | Live Demo**

- Engineered a full-stack, real-time booking platform using **React**, **Node.js**, **PostgreSQL**, and **Redis** to manage **high-concurrency** scenarios and provide a seamless user experience.
- Architected a **WebSocket-driven** live seat map and a **Redis-based locking mechanism** with TTL to prevent concurrent booking conflicts, ensuring data integrity with **atomic PostgreSQL transactions**.
- Automated post-purchase communication by generating **PDF invoices** and dispatching confirmation emails, and **containerized** the full-stack application with a multi-stage **Dockerfile** for streamlined deployment.

Terminal Multiplayer Arcade **GitHub**

- Developed a **Python-based real-time multiplayer game server** supporting 3 different games with **socket-based networking** and **concurrent client handling**.
- Implemented a **lobby system** with **automatic player matching**, comprehensive score tracking, and **persistent game history management**.
- Designed a **plug-and-play game architecture** enabling new games to be added independently without modifying core server logic, paired with an **ANSI-colored terminal UI** for an enhanced user experience.

Skills & Interests

Programming Languages: C/C++, Python, JavaScript, SQL, PHP, MATLAB, HTML/CSS

Frameworks & Libraries: React, Node.js, Express.js, Tailwind CSS

Developer Tools & Platforms: VS Code, Git/GitHub, Linux/Unix, Docker, Redis, PostgreSQL, MongoDB

Security Tools: OpenOCD, GDB, Wireshark, MobSF, Drozer, JADX, JD-GUI, APKTool, ADB, nRF Connect, Bus Pirate, ChipWhisperer, nRF52840 Dongle, Ubertooth One, HackRF One

Embedded, Automotive & Wireless Security: JTAG, SWD, UART, SPI, I2C, CAN, UDS, BLE, GAP/GATT, ECU Diagnostics, PCB Reverse Engineering, Firmware Analysis, Embedded Debugging, Hardware Pentesting

Core Competencies: Data Structures & Algorithms, OOPs, Operating Systems, Computer Networks, Database Management Systems, Embedded Systems, Android Security, RESTful APIs, WebSockets, SDLC, System Design Fundamentals, Agile Methodologies (Scrum, XP), SIEM Platforms

Non-Technical Skills: Adaptation, Problem Solving, Decision-Making, Detail-Oriented, Team Collaboration

Interests: Sports, Music, Technology, Mindful Living

Positions of Responsibility

Open Source Developers Community (OSDC) **August 2023 – July 2025**

Core Team Member

- Contributed across **logistics, technical infrastructure, website development, and judging** for OSDC's flagship hackathons, including **OSDHack** and **RIDEHack**, coordinating with **sponsors** and managing large-scale student participation.
- Co-organized **20+ workshops, speaker sessions, and community events** over two years, driving open-source engagement and collaboration across the JIIT student community.

JIIT Optica Chapter **September 2023 – August 2024**

Technical & Outreach Team Member

- Engineered the front-end for the official Optica chapter website using **ReactJS**, creating a fully responsive and user-centric interface that led to increased member engagement. **Website Link**
- Organized chapter events and workshops while crafting outreach content to promote activities, driving increased student participation and engagement.